Emon - 23701028

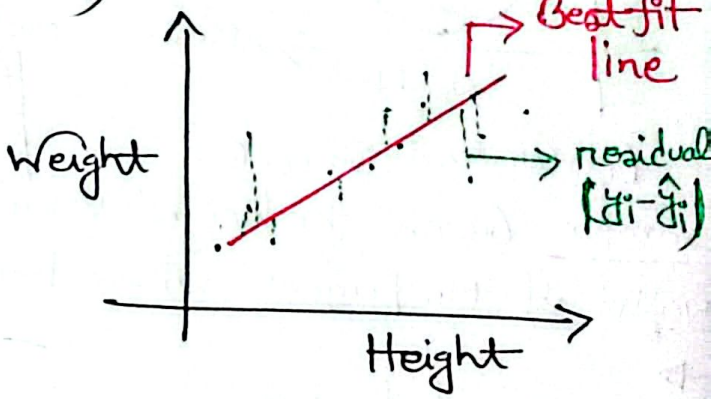
Math Behind Linear Regression

↳ It is a supervised learning algorithm for regression- predicting a continuous number (a price, a temperature, an exam score), as opposed to classification which predicts a category.

*. Vocabulary:

- Feature → the input, denoted as x (Independent Variable) e.g. Height
- Target → the prediction, denoted as y (Dependent) e.g. Weight
- Training example → labled row, written (x^i, y^i) for the ith ex
- Prediction → the model's guess $(\hat{y})$
- Parameters → The knobs that the model tunes. Denoted a $(w \& b)$ or θ_0, θ_1

$$\hat{y} = wx + b$$



↳ For multiple features:

$$\hat{y} = w_1x_1 + w_2x_2 \cdots w_nx_n + b$$

$$\Rightarrow \hat{y} = w^Tx + b$$

Now, how much wrong we are ? ☺

There are so many functions to measure this. Depending on our needs we can introduce our very own function too, but it needs to minimize the error. Some popular functions:

- MSE (Mean Squared Error) $= \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
- MAE (Mean Absolute Error) $= \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$
- RMSE (Root MSE) $= \sqrt{MSE}$
- R^2 (R-Squared) ⇒ It captures the percentage of variance that the model can explain.
  ↳ more এর বেশি.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Now, another major question arises, how do we find the best coefficients (w, b)?

Method 1: (Gradient Descent)

if, $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot x_i$

then,

$$\frac{\partial}{\partial \beta_i}(MSE) = \begin{cases} -\frac{2}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i) ; i=0 \\ -\frac{2}{n}\sum_{i=1}^{n} x_i(y_i - \hat{y}_i) ; i=1 \end{cases}$$

Repeat until converge:

$$= \begin{cases} \beta_0 = \beta_0 - \alpha\left(\frac{\partial}{\partial \beta_i}(MSE)\right) \\ \beta_1 = \beta_1 - \alpha\left(\frac{\partial}{\partial \beta_i}(MSE)\right) \end{cases}$$

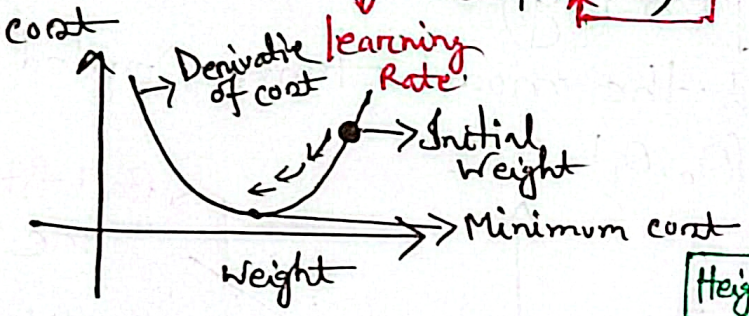


Fig: Gradient Descent.

Method 2: (The Normal Equation)

For Simple Regression,

$$w = \frac{\text{covariance}(x,y)}{\text{variance}(x)}$$

$$\Rightarrow w = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$b = \bar{y} - w\bar{x}$$

For Multiple Regression: → no iteration, no LR

$$\theta = (X^T X)^{-1} \cdot X^T \cdot y$$

→ but costly due to inverse operation

↳ $O(n^3)$

Height(x)	155	160	165	170	175
Weight(y)	52	56	60	63	68

Example: Given the height & weight of several customers, predict the weight of Mr. Perfect, whose height is 161 cm. Using Linear Regression

Step:01 (Compute the means:)

$$\bar{x} = \frac{155+160+165+170+175}{5} = \frac{825}{5} = 165$$

$$\bar{y} = \frac{52+56+60+63+68}{5} = \frac{299}{5} = 59.8$$

Step:02 (Build the Deviation Table)

x_i	155	160	165	170	175	
y_i	52	56	60	63	68	
$x_i - \bar{x}$	-10	-5	0	5	10	
$y_i - \bar{y}$	-7.8	-3.8	0.2	3.2	8.2	
$(x_i - \bar{x})(y_i - \bar{y})$	78.0	19.0	0	16.0	82.0	195.0
$(x_i - \bar{x})^2$	100	25	0	25	100	250

Step:03 (Compute w & b)

$$w = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{195}{250} = 0.78$$

$$b = \bar{y} - w\bar{x} = -68.9$$

Step:04 (Write the Regression Line)

$$\hat{y} = 0.78x - 68.9$$

Step:05 (Predict) $\Rightarrow x = 161$

$$\hat{y} = (0.78)(161) + (-68.9) = 56.68$$